

LABSHEET - 3**HIMANI CHAUDHARY****BTECH(CSE) 3RD YEAR****CU240250967****SECTION - B****ROLL NO. - 24****Mastery in Data Analytics and Visualizations and Career Advancement (Technical)**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
import statsmodels.api as sm
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
```

```
# =====
# 1. LOAD DATASET
# =====

print("\n" + "=" * 60)
print("1. LOADING TITANIC DATASET")
print("=" * 60)

try:
    df = sns.load_dataset("titanic")
except Exception:
    print("Unable to download Titanic dataset.")
    exit()

print("\nFirst 5 rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())
print("Himani Chaudhary")
print("CU240250967")
```

```
=====
1. LOADING TITANIC DATASET
=====
```

First 5 rows:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
0	0	3	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	1	3	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	0	3	male	35.0	0	0	8.0500	S	Third	

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

Dataset Shape:
(891, 15)

Column Names:

```
['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare', 'embarked', 'class', 'who', 'adult_male', 'deck', 'embark_tow']
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```

```
# 2. BASIC INFORMATION
# =====

print("\n" + "=" * 60)
```

```

print("2. BASIC INFORMATION")
print("=" * 60)

print("\nDataset Information:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())
print("Himani Chaudhary")
print("CU240250967")

```

2. BASIC INFORMATION

```

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column             Non-Null Count  Dtype
---  -
0   survived           891 non-null   int64
1   pclass             891 non-null   int64
2   sex                891 non-null   object
3   age                714 non-null   float64
4   sibsp             891 non-null   int64
5   parch             891 non-null   int64
6   fare              891 non-null   float64
7   embarked          889 non-null   object
8   class             891 non-null   category
9   who               891 non-null   object
10  adult_male        891 non-null   bool
11  deck              203 non-null   category
12  embark_town       889 non-null   object
13  alive             891 non-null   object
14  alone             891 non-null   bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None

```

```

Missing Values:
survived      0
pclass        0
sex           0
age           177
sibsp         0
parch         0
fare          0
embarked      2
class         0
who           0
adult_male    0
deck          688
embark_town   2
alive         0
alone         0
dtype: int64
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```

3. DESCRIPTIVE STATISTICS

```

# =====

print("\n" + "=" * 60)
print("3. DESCRIPTIVE STATISTICS")
print("=" * 60)

# Numerical columns
numeric_columns = ["age", "fare", "sibsp", "parch"]

print("\nMean:")
print(df[numeric_columns].mean())

print("\nMedian:")
print(df[numeric_columns].median())

print("\nMode:")
print(df[numeric_columns].mode().iloc[0])

print("\nVariance:")
print(df[numeric_columns].var())

print("\nStandard Deviation:")
print(df[numeric_columns].std())

```

```
print("\nComplete Descriptive Statistics:")
print(df[numeric_columns].describe())
print("Himani Chaudhary")
print("CU240250967")
```

```
=====
3. DESCRIPTIVE STATISTICS
=====
```

```
Mean:
age      29.699118
fare     32.204208
sibsp    0.523008
parch    0.381594
dtype: float64
```

```
Median:
age      28.0000
fare     14.4542
sibsp    0.0000
parch    0.0000
dtype: float64
```

```
Mode:
age      24.00
fare      8.05
sibsp    0.00
parch    0.00
Name: 0, dtype: float64
```

```
Variance:
age      211.019125
fare     2469.436846
sibsp     1.216043
parch     0.649728
dtype: float64
```

```
Standard Deviation:
age      14.526497
fare     49.693429
sibsp     1.102743
parch     0.806057
dtype: float64
```

```
Complete Descriptive Statistics:
      age      fare      sibsp      parch
count  714.000000  891.000000  891.000000  891.000000
mean    29.699118  32.204208    0.523008    0.381594
std     14.526497  49.693429    1.102743    0.806057
min      0.420000    0.000000    0.000000    0.000000
25%     20.125000    7.910400    0.000000    0.000000
50%     28.000000   14.454200    0.000000    0.000000
75%     38.000000   31.000000    1.000000    0.000000
max      80.000000  512.329200    8.000000    6.000000
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```

```
# 4. PEARSON CORRELATION
# =====
```

```
print("\n" + "=" * 60)
print("4. PEARSON CORRELATION ANALYSIS")
print("=" * 60)
```

```
correlation = df[numeric_columns].corr(method="pearson")
```

```
print("\nCorrelation Matrix:")
print(correlation)
```

```
# Correlation heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)
```

```
plt.title("Pearson Correlation Matrix")
plt.tight_layout()
plt.show()
print("Himani Chaudhary")
print("CU240250967")
```

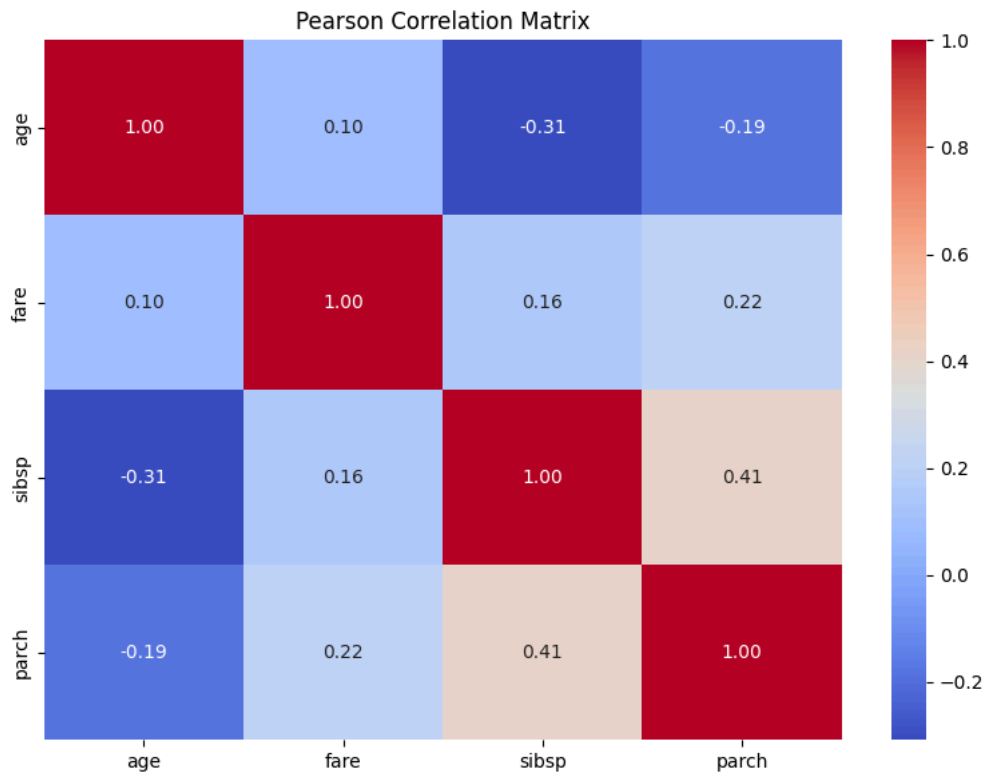
4. PEARSON CORRELATION ANALYSIS

Correlation Matrix:

```

age      fare      sibsp      parch
age      1.000000  0.096067 -0.308247 -0.189119
fare      0.096067  1.000000  0.159651  0.216225
sibsp     -0.308247  0.159651  1.000000  0.414838
parch     -0.189119  0.216225  0.414838  1.000000

```



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5. HYPOTHESIS TESTING

Independent Sample T-Test

=====

```

print("\n" + "=" * 60)
print("5. INDEPENDENT SAMPLE T-TEST")
print("=" * 60)

# Remove missing values
male_fare = df[df["sex"] == "male"]["fare"].dropna()
female_fare = df[df["sex"] == "female"]["fare"].dropna()

# H0: There is no significant difference in average fare
# H1: There is a significant difference in average fare

t_stat, p_value = stats.ttest_ind(
    male_fare,
    female_fare,
    equal_var=False
)

print("\nNull Hypothesis (H0):")
print("There is no significant difference between male and female average fare.")

print("\nAlternative Hypothesis (H1):")
print("There is a significant difference between male and female average fare.")

print("\nMale Average Fare:", male_fare.mean())
print("Female Average Fare:", female_fare.mean())

print("\nT-statistic:", t_stat)
print("P-value:", p_value)

alpha = 0.05

if p_value < alpha:
    print("\nResult: Reject H0.")

```

```

    print("There is a statistically significant difference.")
else:
    print("\nResult: Fail to Reject H0.")
    print("There is no statistically significant difference.")
    print("Himani Chaudhary")
    print("CU240250967")

```

```

=====
5. INDEPENDENT SAMPLE T-TEST
=====

```

Null Hypothesis (H0):
There is no significant difference between male and female average fare.

Alternative Hypothesis (H1):
There is a significant difference between male and female average fare.

Male Average Fare: 25.523893414211443
Female Average Fare: 44.47981783439491

T-statistic: -5.077499013458909
P-value: 5.388057409693902e-07

Result: Reject H0.
There is a statistically significant difference.

```

# 6. ONE-WAY ANOVA
# =====

```

```

print("\n" + "=" * 60)
print("6. ONE-WAY ANOVA")
print("=" * 60)

```

Compare fares among passenger classes

```

class_1 = df[df["pclass"] == 1]["fare"].dropna()
class_2 = df[df["pclass"] == 2]["fare"].dropna()
class_3 = df[df["pclass"] == 3]["fare"].dropna()

```

```

f_stat, anova_p = stats.f_oneway(
    class_1,
    class_2,
    class_3
)

```

```

print("\nNull Hypothesis (H0):")
print("There is no significant difference among the mean fares of the three classes.")

```

```

print("\nAlternative Hypothesis (H1):")
print("At least one class has a significantly different mean fare.")

```

```

print("\nClass 1 Mean Fare:", class_1.mean())
print("Class 2 Mean Fare:", class_2.mean())
print("Class 3 Mean Fare:", class_3.mean())

```

```

print("\nF-statistic:", f_stat)
print("P-value:", anova_p)

```

```

if anova_p < alpha:
    print("\nResult: Reject H0.")
    print("There is a statistically significant difference among groups.")
else:
    print("\nResult: Fail to Reject H0.")
    print("There is no statistically significant difference among groups.")
    print("Himani Chaudhary")
    print("CU240250967")

```

```

=====
6. ONE-WAY ANOVA
=====

```

Null Hypothesis (H0):
There is no significant difference among the mean fares of the three classes.

Alternative Hypothesis (H1):
At least one class has a significantly different mean fare.

Class 1 Mean Fare: 84.1546875
Class 2 Mean Fare: 20.662183152173913
Class 3 Mean Fare: 13.675550101832993

F-statistic: 242.34415651744814
P-value: 1.0313763209141171e-84

Result: Reject H_0 .
There is a statistically significant difference among groups.

```
# 7. SIMPLE LINEAR REGRESSION
# =====

print("\n" + "=" * 60)
print("7. SIMPLE LINEAR REGRESSION")
print("=" * 60)

# Relationship between Age and Fare

regression_data = df[["age", "fare"]].dropna()

X = regression_data[["age"]]
y = regression_data["fare"]

model = LinearRegression()

model.fit(X, y)

y_pred = model.predict(X)

slope = model.coef_[0]
intercept = model.intercept_

r2 = r2_score(y, y_pred)

print("\nRegression Equation:")
print(f"Fare = {intercept:.2f} + ({slope:.2f} × Age)")

print("\nRegression Coefficient:", slope)
print("Intercept:", intercept)
print("R² Score:", r2)
print("Himani Chaudhary")
print("CU240250967")
```

```
=====
7. SIMPLE LINEAR REGRESSION
=====

Regression Equation:
Fare = 24.30 + (0.35 × Age)

Regression Coefficient: 0.34996368174402326
Intercept: 24.300901449288645
R² Score: 0.009228809267447624
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```

```
# 8. REGRESSION PLOT
# =====

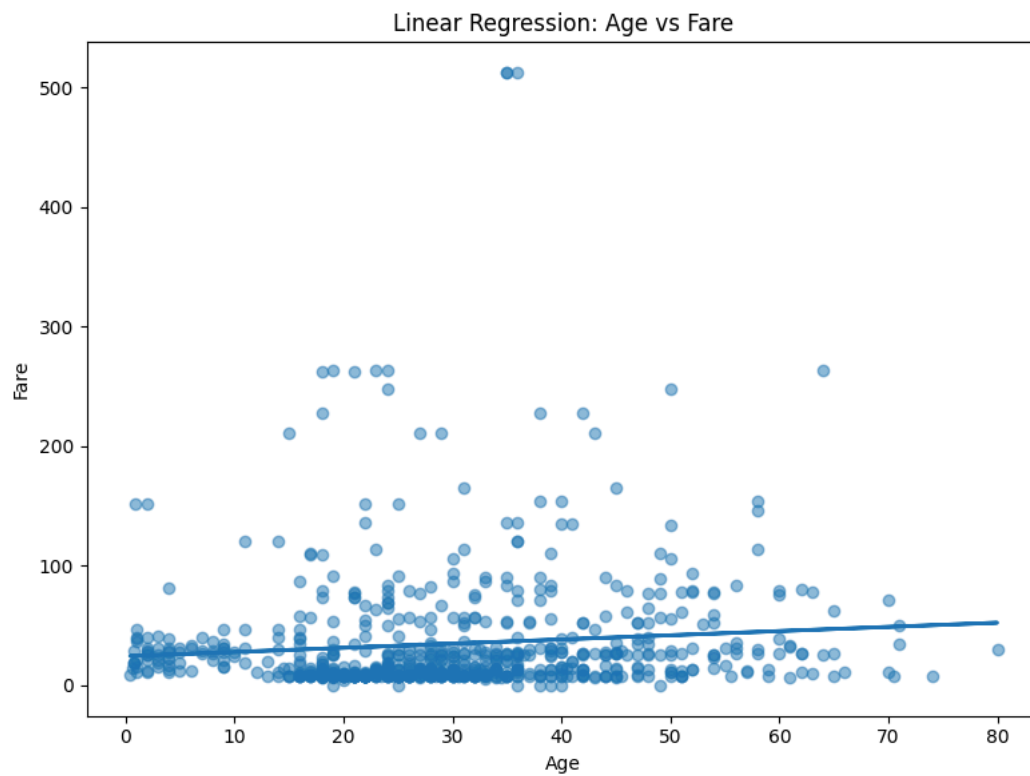
plt.figure(figsize=(8, 6))

plt.scatter(
    regression_data["age"],
    regression_data["fare"],
    alpha=0.5
)

plt.plot(
    regression_data["age"],
    y_pred,
    linewidth=2
)

plt.xlabel("Age")
plt.ylabel("Fare")
plt.title("Linear Regression: Age vs Fare")

plt.tight_layout()
plt.show()
print("Himani Chaudhary")
print("CU240250967")
```



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```
# 9. CONFIDENCE INTERVAL
# =====

print("\n" + "=" * 60)
print("9. CONFIDENCE INTERVAL")
print("=" * 60)

# OLS regression for confidence interval

X_stats = sm.add_constant(X)

ols_model = sm.OLS(y, X_stats).fit()

print("\nOLS Regression Summary:")
print(ols_model.summary())

print("\n95% Confidence Intervals:")
print(ols_model.conf_int())
print("Himani Chaudhary")
print("CU240250967")
```

```
=====
9. CONFIDENCE INTERVAL
=====

OLS Regression Summary:
                        OLS Regression Results
=====
Dep. Variable:          fare      R-squared:                0.009
Model:                  OLS      Adj. R-squared:            0.008
Method:                 Least Squares      F-statistic:         6.632
Date:                   Wed, 26 Aug 2026    Prob (F-statistic):    0.0102
Time:                   13:37:57           Log-Likelihood:       -3843.0
No. Observations:       714              AIC:                7690.
Df Residuals:           712              BIC:                7699.
Df Model:                1
Covariance Type:        nonrobust
=====
                        coef      std err          t      P>|t|      [0.025      0.975]
-----
const                24.3009      4.492        5.410      0.000      15.481     33.120
age                   0.3500      0.136        2.575      0.010       0.083     0.617
=====
Omnibus:              716.815    Durbin-Watson:        1.925
Prob(Omnibus):         0.000    Jarque-Bera (JB):     30883.661
Skew:                  4.654    Prob(JB):              0.00
Kurtosis:              33.846    Cond. No.              75.3
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

95% Confidence Intervals:

	0	1
const	15.481351	33.120452
age	0.083165	0.616762

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10. FINAL SUMMARY

=====

```
print("\n" + "=" * 60)
print("10. FINAL SUMMARY")
print("=" * 60)
```

```
print("""
Statistical analysis was performed on the Titanic dataset.
```

1. Descriptive statistics were calculated using mean, median, mode, variance and standard deviation.
2. Pearson correlation was used to find relationships between numerical variables.
3. Independent Sample T-Test was used to compare the average fare between male and female passengers.
4. One-Way ANOVA was used to compare fares among passenger classes.
5. Linear Regression was used to study the relationship between age and fare.
6. P-values were used to determine statistical significance.
7. R^2 score was used to measure how well the regression model explains the variation in the dependent variable.

Statistical analysis helps in making data-driven decisions and understanding relationships between variables.

```
""")
print("Himani Chaudhary")
print("CU240250967")
```

```
=====
10. FINAL SUMMARY
=====
```

Statistical analysis was performed on the Titanic dataset.

1. Descriptive statistics were calculated using mean, median, mode, variance and standard deviation.
2. Pearson correlation was used to find relationships between numerical variables.
3. Independent Sample T-Test was used to compare the average fare between male and female passengers.
4. One-Way ANOVA was used to compare fares among passenger classes.
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7. R^2 score was used to measure how well the regression model explains the variation in the dependent variable.

Statistical analysis helps in making data-driven decisions and understanding relationships between variables.

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CU240250967

END OF EXPERIMENT

=====

```
print("\n" + "=" * 60)
print("EXPERIMENT COMPLETED SUCCESSFULLY")
print("=" * 60)
print("Himani Chaudhary")
print("CU240250967")
```

```
=====
EXPERIMENT COMPLETED SUCCESSFULLY
=====
Himani Chaudhary
CU240250967
```